Divyesh Burande

Edinburgh, Scotland • Mail • Github • LinkedIn

Education

Heriot-Watt University — Edinburgh, Scotland
MSc Data Science - Merit — June 2026

- Dissertation: Evaluating Rust vs Python for Audio Feature Extraction and Classification: A Benchmarking Study - 85/100.

Technical Skills

Programming & Data: Python, SQL, Rust, Pandas, NumPy, SciPy
Machine Learning: scikit-learn, PyTorch, Regression, Classification, Feature Engineering, Model Selection & Evaluation, Cross-Validation
Statistics & Experimentation: Hypothesis Testing, Experimental Design, Regression Analysis, ANOVA, Statistical Modelling
Data Visualisation: Matplotlib, Seaborn, Exploratory Data Analysis
Tools: Git, GitHub, Jupyter, Linux

Projects

Chest X-Ray Multi-Class Classification
Python | PyTorch | CNN | ResNet18 | Transfer Learning

- Developed a four-class image classification pipeline to distinguish COVID-19, normal, pneumonia and tuberculosis chest X-rays.
- Built a CNN baseline from scratch (0.829 macro F1) and applied ResNet18 transfer learning with class weighting and partial fine-tuning, improving macro F1 to 0.881.
- Conducted per-class error analysis, identifying 0.547 recall for the NORMAL class and examining prediction bias associated with class imbalance.

Particle Swarm Optimisation for Neural Network Training
Python | Neural Networks | Particle Swarm Optimisation | Regression

- Implemented a feed-forward neural network and Particle Swarm Optimisation algorithm from scratch, without deep-learning frameworks, for concrete-strength regression.
- Designed controlled experiments across network architectures, swarm configurations and optimisation coefficients, reducing test MAE from 29.63 to 10.41 during architecture experiments.
- Analysed exploration–exploitation trade-offs under fixed computational budgets; the selected final model achieved 11.67 MAE and 213.45 MSE.

Python vs Rust for Audio Feature Extraction and Classification
MSc Dissertation | Python | Rust | scikit-learn | Statistical Analysis

- Designed a 2×2 experimental study comparing Python and Rust implementations of an audio feature-extraction and KNN classification pipeline using the GTZAN dataset.
- Conducted 1,000 benchmark runs across four implementations and five dataset sizes, evaluating runtime, peak memory, CPU utilisation and classification accuracy using stratified 5-fold cross-validation.
- Applied ANOVA and post-hoc statistical analysis, identifying significant Python–Rust differences in computational performance without significant differences in classification accuracy.

Experience

Cambridge Wealth Pune, India
Business Analyst Intern Feb 2024 – Sep 2024

- Developed automated reporting workflows using Python and Excel, generating portfolio summaries, capital-gains reports and transaction analyses for 10+ clients.
- Queried and analysed structured financial datasets using SQL to support reporting and internal decision-making.
- Built and maintained dashboards to improve visibility into portfolio performance and operational processes.

OnlineSales.ai Pune, India
Product Analyst Intern Jan 2023 – Jul 2023

- Analysed campaign and customer-engagement data using Excel and internal analytics platforms to support performance tracking and optimisation decisions.
- Evaluated digital advertising and video platforms through structured feature and usability comparisons.
- Collaborated with cross-functional teams to interpret performance metrics and support data-driven decision-making.

Additional UK Experience

Sainsbury's Edinburgh, UK
Local Assistant Jun 2026 – Present

- Support customer service, tills, self-checkout, stock replenishment, and online-order operations in a fast-paced retail environment.

William Hill Edinburgh, UK
Customer Service Representative Jan 2025 – Present

- Handle real-time transactions and customer queries in a regulated environment, maintaining accuracy and compliance with internal procedures.